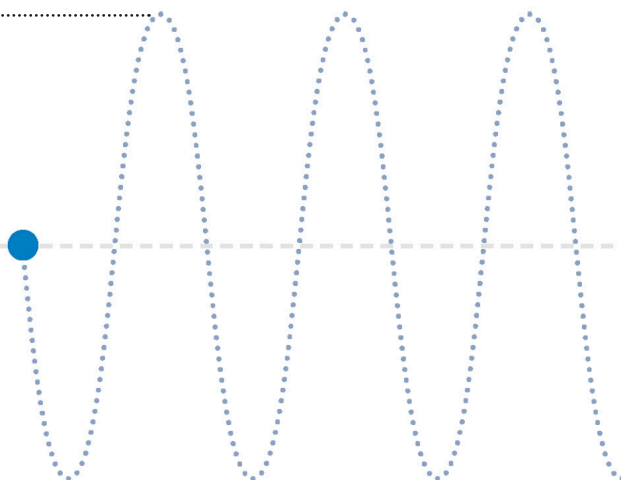


APPROACHING A BARRIER

In terms of classical physics, the forces binding an atomic nucleus create an energy barrier or "potential well." Classically, a particle could only escape a well if it had enough kinetic energy to overcome the potential energy deficit. Energy below this value would mean that the particle was trapped forever.



WHAT BARRIER?

An effect known as tunneling explains how subatomic particles can sometimes cross apparently insurmountable barriers. During radioactive alpha decay (see p.113), for example, a cluster of protons and neutrons spontaneously breaks free from a larger atomic nucleus by overcoming the binding energy that holds the nucleus together. Such a leap is impossible in classical physics, but the edges of a quantum wave function can reach beyond the barrier, allowing for a small chance of the particle being found there.